

# Optimised blind image watermarking method based on firefly algorithm in DWT-QR transform domain

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**Abstract:** Firefly algorithm (FA) is one of the newly developed nature inspired optimisation algorithm, inspired by the flashing behaviour of fireflies that a firefly tends to be attracted towards other fireflies with higher brightness. Thus FA has two advantages: local attractions and automatic regrouping. Based on these good properties, a novel image watermarking method based on FA in discrete wavelet transform (DWT)-QR transform domain is proposed in this study. Structural similarity index measure and bit error rate are used in the objective function to trade-off invisibility and robustness. The experiment results show that the proposed image watermarking method not only meet the need of invisibility, but also has better or comparable robustness as compared with some related methods.

## 1 Introduction

Nowadays, the transportation of digital images are growing rapidly through internet and thus the content protection of image has become a crucial point. This specific need motivates more and more researchers to develop efficient methods in content protection filed, such as steganography, digital watermarking and visual cryptography. As one of the popular and efficient method, the digital image watermarking, embed the watermark into the cover image, which gives a digital signature of the author. Numerous efforts have been devoted to the development of image watermarking method [1–5].

Invisibility and robustness are the main requirements for developing a new image watermarking method. The image watermarking method can be classified into two categories, i.e. spatial domain method and transform domain method. The transform domain image watermarking method transforms the image into the transform domain and then modifies the transform coefficients to embed watermark. Currently, the discrete cosine transform (DCT) [6], the discrete Fourier transform [7], matrix decomposition [8–10], the discrete wavelet transform (DWT) [11, 12] and the discrete fractional Fourier transform [13, 14] are frequently used in transform domain image watermarking techniques, where common matrix decomposition include singular value decomposition (SVD), QR decomposition and Schur decomposition. Each transform and decomposition technique has its own advantages and disadvantages for digital image watermarking.

Since combined transforms may compensate for the drawbacks of each other, the combination of two or more transformations can further improve the performance of watermarking method. The DWT based image watermarking techniques have advantages such as multi-resolution representation and better energy compression, hence it can provide high robustness to image processing attacks but low robustness to some geometric attacks. The matrix decomposition can be used to extract the geometric features of an image, which is more robust against geometric attacks. Therefore, the image watermarking method based on combined DWT and matrix decomposition take the advantages of DWT and matrix decomposition and thus robust against both geometric attacks and image processing attacks [2, 15–21]. These traditional image watermarking methods have advantages in terms of high

computing speed, but they cannot balance the invisibility and robustness automatically because invisibility and robustness are in conflict to each other. The tradeoff between invisibility and robustness can be viewed as an optimisation problem. Nature inspired algorithms (NIAs) have been employed to solve the optimisation problem, such as genetic algorithm (GA) [22], particle swarm optimisation (PSO) [23, 24], differential evolution [25, 26], artificial bee colony (ABC) [27, 28], ant colony optimisation [29]. Existing papers show that NIAs can improve the performances of image watermarking method. More information of optimised image watermarking method based on NIAs can be found in [30, 31].

Recently, a new optimisation algorithm inspired by firefly known as firefly algorithm (FA) is developed by Yang [32]. This algorithm is different from other counterparts such as GA, PSO and few other NIAs. Many researchers have employed FA to different fields successfully and found its advantages compared with other NIAs [33–37]. Zhang *et al.* [33] has applied FA to image registration and showed that FA based method can achieve the closest solution to the actual spatial transformation parameters compared with GA, PSO and ABC. Hassanzadeh *et al.* [34] has applied FA to image segmentation and the experimental results show that FA-based method is far more efficient than Otsu's method and recursive Otsu. Kanimozhi and Latha [35] has applied FA to intelligent image retrieval and the retrieval performance shows that FA-based method yields higher average precision and recall when compared with the existing methods such as PSO, GA, support vector machine and query point movement. Ali and Ahn [38] and Mishra *et al.* [39] have made some exploratory research about the application of FA in image watermarking. Hence it becomes necessary to detect more potential advantages of FA in image watermarking.

To the best of our knowledge, most of the papers focused on SVD-based image watermarking. Few research papers are associated with the image watermarking based on QR decomposition and it has been proved that the image watermarking method based on QR decomposition offers better or at least comparable performance than SVD and DCT in terms of some performance metrics [10]. Therefore, based on the above two reasons, it is interesting and worth noting to investigate the image watermarking method based on FA in DWT-QR domain. The remaining sections of this paper are organised as follows. In

Section 2, we review some basic concepts about DWT, QR decomposition and FA. In Section 3, a novel image watermarking method is proposed based on the FA in DWT-QR domain. The performance analysis of this novel image watermarking method is shown in Section 4. Finally, conclusions are stated in Section 5.

## 2 Preliminary

### 2.1 Discrete wavelet transform

DWT is one of the valuable mathematical transform having numerous applications in science, engineering and computer science. More particular, it is used for JPEG2000 image compression. Due to its advantages such as similarity of data structure with respect to the resolution and available decomposition at any level, DWT has been successfully applied to image watermarking field. Each level of DWT decompose an image into four subbands namely a lower resolution approximation component LL, horizontal detail component HL, vertical detail component LH and diagonal detail component HH. Most of the information contained in the original image is concentrated into the LL subband after one level DWT. The LL subband can further be decomposed to obtain another level of decomposition. The decomposition continues on the LL subband until the desired result is reached.

### 2.2 QR decomposition

QR decomposition is one of the important matrix decomposition. It can be used in linear least squares problem and principal component analysis. There are several methods for computing the QR decomposition, such as by means of the Gram–Schmidt process, Householder transformations and Givens rotations.

Any matrix  $A$  of size  $m \times n$  can be decomposed as following:

$$A = QR \quad (1)$$

where  $Q$  is a  $m \times n$  orthogonal matrix, its column vectors are orthogonal unit vector.  $R$  is an  $n \times n$  upper triangular matrix. If  $A$  is non-singular, then this decomposition is unique.

In [10], it has been proved that  $|R(1, :)|$  are larger than  $|R(i, :)|$  ( $i = 2, 3, \dots, n$ ) if the columns of  $A$  are correlated, where the  $i$ th row of  $R$  is denoted as  $R(i, :)$ . Therefore, if  $A$  is the original cover image, small change of  $R(1, :)$  will not lead to image distortion. That is the reason why we embed watermark into  $R(1, :)$  in this paper. The other feature of QR decomposition is that the size of  $A$  given in (1) is flexible and it is not bound to be a square matrix.

### 2.3 Firefly algorithm

FA is a nature-inspired optimisation algorithm. It was proposed by Yang [32] and inspired by the flashing behaviour of fireflies, which act as a signal system to attract other fireflies. There are three assumptions in FA:

- All fireflies are unisexual, so that any individual firefly can be attracted to all other fireflies.
- Attractiveness is proportional to their brightness and both attractiveness and brightness decrease with increasing of their mutual distance. It means that for any two fireflies, the less bright firefly will be attracted towards the brighter firefly. If no firefly is brighter than a given firefly, then it will move randomly.
- The brightness of a firefly is determined by the objective function. For simplicity, the brightness of a firefly at a particular location is equal to the objective function value.

As mentioned above, FA includes two basic factors: brightness and attractiveness. The attractiveness of a firefly is determined by its brightness. The attractiveness of a firefly is inversely proportional to the distance between the two fireflies. Therefore, the attractiveness of a firefly can be defined as following:

$$\beta = \beta_0 e^{-\gamma r_{ij}^2} \quad (2)$$

where  $r_{ij}$  is the distance between two fireflies  $i$  and  $j$ ,  $\beta_0$  is the attractiveness at  $r = 0$  and  $\gamma$  is the light absorption coefficient of the medium.

The distance between any two fireflies is defined as the Cartesian distance. It is worth noting that the distance  $r$  is not limited to the Euclidean distance, any measure that can effectively characterise the quantities of interest in the optimisation problem can be used to define the distance  $r$  [32].

For any given two populations of fireflies  $x_i$  and  $x_j$ , if  $f(x_j) > f(x_i)$ , move firefly  $i$  towards  $j$  according to the following equation

$$x_i^{t+1} = x_i^t + \beta(x_j^t - x_i^t) + \alpha \epsilon \quad (3)$$

where, the second term is due to the attraction, the third term is disturbance term which set for avoiding premature fall into local optimum.  $\alpha$  is a parameter controlling the step size and  $\epsilon$  is a vector drawn from a Gaussian or other distribution.

Equation (3) is random walk biased towards the brighter fireflies. If  $\beta_0 = 0$ , it becomes a simple random walk. For most implement, we can take  $\beta_0 = 1$ ,  $\alpha \in [0, 1]$ . The values of parameter  $\gamma$  are crucially important in determining the speed of convergence and how the FA behaves. In theory,  $\gamma \in [0, +\infty)$ . It can be shown that the limiting case  $\gamma \rightarrow 0$  corresponds to the standard PSO. However, in practice,  $\gamma = O(1)$  is determined by the characteristic of the system to be optimised. Therefore,  $\gamma$  is selected from 0.1 to 10 in most applications [32].

The algorithm of FA can be summarised as follows:

Step 1: Input initial parameters of FA and the objective function  $f$ . The initialisation parameters include  $n$  number of fireflies, maximum attractiveness  $\beta_0$ , step size  $\alpha$ , light absorption coefficient  $\gamma$  and maximum number of iterations  $T$ .

Step 2: Initialise a population of  $n$  fireflies  $x_i$  ( $i = 1, 2, \dots, n$ ).

Step 3: Compute the brightness of each firefly according to the objective function, and the attractiveness of each firefly to other fireflies according to (2).

Step 4: If  $f(x_j) > f(x_i)$ , move firefly  $i$  towards  $j$  according to (3), then update the location of each firefly.

Step 5: Recalculate the brightness of each firefly according to the updated location obtained in step 4.

Step 6: If the  $T$  number of iterations are reached, the global best is obtained as an output. Else, go to step 3 and do the next iteration until maximum number of iteration  $T$  is reached.

## 3 Proposed algorithm

### 3.1 Watermark embedding

In this subsection, the watermark embedding procedure of proposed method is introduced. Without loss of generality, assume the size of cover image is  $M \times M$ , the size of watermark image is  $N \times N$  satisfying  $N = M/8$ . The detailed procedure is elaborated as follows:

- The cover image  $X$  is converted into a column vector  $Y$  of size  $M^2 \times 1$ .
- Sort the elements of  $Y$  in ascending order using  $[Z, P] = \text{sort}(Y)$ , where 'sort' is a function of matlab,  $Z$  is the sorting result of  $Y$  and  $P$  records the original positions of these elements of  $Z$ . Mathematically,  $Y(i, 1) = Z(P(i, 1), 1)$ .
- Reshape  $Z$  back to scrambled cover image of size  $M \times M$ , denoted as  $X'$ .
- Decompose  $X'$  using one level DWT to obtain four subbands  $LL$ ,  $HL$ ,  $LH$  and  $HH$ , where the approximate subband  $LL$  of size  $M/2 \times M/2$  is used to embed watermark in this paper. In this paper, the 'db1' wavelet is used in the DWT.

- v. Divide  $LL$  into non-overlapping blocks  $\{a_{ij}\} (1 \leq i, j \leq N)$ , where the size of each block is  $4 \times 4$  and the total number of blocks are  $N^2$ .
- vi. Apply QR decomposition to each block to obtain two components  $Q_{ij} (1 \leq i, j \leq N)$  and  $R_{ij} (1 \leq i, j \leq N)$  as follows:

$$[Q_{ij}, R_{ij}] = qr(a_{ij}) \quad (1 \leq i, j \leq N)$$

- vii The first row of each  $R_{ij} (1 \leq i, j \leq N)$  is selected to embed binary watermark using the formula given by:

$$R_{ij}^w(1, :) = \begin{cases} R_{ij}(1, :) + \lambda \cdot \mathbf{K} & \text{if } w_{ij} = 1 \\ R_{ij}(1, :) - \lambda \cdot \mathbf{K} & \text{if } w_{ij} = 0 \end{cases} \quad (4)$$

where  $\mathbf{K}$  is a  $1 \times 4$  dimensional random integral vector, whose values are independent and uniformly distributed over the interval  $[-1, 1]$ .  $R_{ij}(1, :)$  denotes the first row of  $R_{ij}$ ,  $R_{ij}^w$  denotes the watermarked  $R_{ij}$ .  $\lambda$  is watermark embedding strength obtained by FA and it will be introduced in detail in Section 3.3.

- viii Each watermarked block is obtained by  $b_{ij}^w = Q_{ij} R_{ij}^w$ .
- ix. Merge all these watermarked blocks together to obtain  $LL^w$  using operation inverse to step 5, where  $LL^w$  denotes the watermarked approximate subband.
- x. Perform IDWT to  $LL^w$  and other three subbands  $HL$ ,  $LH$  and  $HH$  obtained in step 4, the watermarked scrambled cover image  $\{X'\}^w$  can be obtained.
- xi. Use the position vector  $\mathbf{P}$  obtained in step 2 to obtain the watermarked cover image  $X^w$ .

### 3.2 Watermark extraction

In this subsection, the watermark extraction procedure is shown. The position vector  $\mathbf{P}$  and watermarked cover image  $X^w$  are used in watermark extraction. It is worth to notice that the using of position vector  $\mathbf{P}$  can not only improve the security of watermark extraction as a secret key, but also decrease the bit error rate (BER) between original watermark and extracted watermark. The detailed steps of the watermark extraction are described as follows:

- i. Reshape the watermarked cover image  $X^w$  back to the watermarked scrambled cover image  $\{X'\}^w$  using the position vector  $\mathbf{P}$ .
- ii. Apply one level DWT to the watermarked scrambled cover image  $\{X'\}^w$  to obtain four subbands  $LL'$ ,  $HL'$ ,  $LH'$  and  $HH'$ .  $LL'$  is used to extract the watermark.
- iii.  $LL'$  is divided into non-overlapping blocks  $b_{ij} (1 \leq i, j \leq N)$ , where the size of each block is  $4 \times 4$  and the total number of blocks are  $N^2$ .
- iv. Apply QR decomposition to each block  $b_{ij} (1 \leq i, j \leq N)$  to obtain two components  $Q'_{ij}$  and  $R'_{ij}$ .
- v. The first row of  $R'_{ij}$  is used to extract the watermark using the formula given by:

$$w'(i, j) = \begin{cases} 1, & \text{if } \text{corrcoef}(R'_{ij}(1, :), \mathbf{K}) \geq 0 \\ 0, & \text{if } \text{corrcoef}(R'_{ij}(1, :), \mathbf{K}) < 0 \end{cases} \quad (5)$$

where  $R'_{ij}(1, :)$  denotes the first row of  $R'_{ij}$ . The  $\text{corrcoef}(R'_{ij}(1, :), \mathbf{K})$  is the standard correlation coefficient, which is directly determined by the variance of  $R'_{ij}(1, :)$ .

### 3.3 Application of the FA algorithm in finding optimal parameter

Invisibility and robustness are the main points that are considered by many researchers in digital image watermarking techniques.

However, invisibility and robustness are interrelated and contradictory. Therefore, the problem of how to tradeoff invisibility and robustness is important to meet the needs of applications. In recent years, NIAs have been applied in the image watermarking field as a powerful tool, which can effectively solve the problem of tradeoff between invisibility and robustness. Generally, watermark embedding strength and watermark embedding position are considered as two important factors in the objective function of NIAs. In this paper, FA is employed to search for optimal watermark embedding strength in order to achieve the optimal performance of the proposed image watermarking method.

Invisibility is frequently measured by the performance metrics such as peak signal to noise ratio (PSNR), the normalised cross-correlation (NC), structural similarity index measure (SSIM) and correlation coefficient (Corr). Robustness is frequently measured by NC, SSIM and BER. PSNR is commonly used as an index to measure the quality of the reconstructed watermarked image. NC, SSIM and BER are used to measure the similarity index between the two images. Mathematically, these performance metrics are defined as:

$$\text{PSNR}(X, Y) = 10 \log_{10} \frac{MN \cdot X_{\max}^2}{\sum_{m=1}^M \sum_{n=1}^N [X(m, n) - Y(m, n)]^2} \quad (6)$$

$$\begin{aligned} \text{NC}(X, Y) &= \frac{\sum_{m=1}^M \sum_{n=1}^N X(m, n) \cdot Y(m, n)}{\sqrt{\sum_{m=1}^M \sum_{n=1}^N [X(m, n)]^2} \cdot \sqrt{\sum_{m=1}^M \sum_{n=1}^N [Y(m, n)]^2}} \end{aligned} \quad (7)$$

where  $X$  and  $Y$  are two images of same size  $M \times N$ .  $X_{\max}$  is the maximum pixel value of the image  $X$ .

$$\text{SSIM}(X, Y) = \frac{(\mu_X \mu_Y + c_1)(2\sigma_{XY} + c_2)}{(\mu_X^2 + \mu_Y^2 + c_1)(\sigma_X^2 + \sigma_Y^2 + c_2)} \quad (8)$$

where  $\mu_X$  and  $\mu_Y$  is the average of image  $X$  and  $Y$ , respectively.  $\sigma_X$  and  $\sigma_Y$  is the variance of image  $X$  and  $Y$ , respectively.  $\sigma_{XY}$  is the covariance of image  $X$  and  $Y$ , respectively, and  $c_1$  and  $c_2$  are two variables to stabilise the division with a weak denominator.

$$\text{BER}(X, Y) = \frac{\sum_{m=1}^M \sum_{n=1}^N X(m, n) \oplus Y(m, n)}{M \times N} \quad (9)$$

where  $\oplus$  denotes the exclusive-OR(XOR) operation.

In optimised digital image watermarking, many researchers have defined various objective functions to tradeoff watermark invisibility and robustness. In all objective functions, PSNR, NC and Corr are commonly used to measure watermark invisibility, such as the objective functions given by (10)–(13). SSIM is relatively less used than the above three metrics, such as the objective functions given by (14). BER and NC are commonly used to measure the robustness, where BER describes the error more intuitively.

$$f = \text{PSNR}(X, X^w) + 30 \cdot \sum_{i=1}^N \text{BER}(w, w'_i) \quad (10)$$

$$f = \text{PSNR}(X, X^w)/100 + \sum_{i=1}^N \text{NC}(w, w'_i) \quad (11)$$

$$f = \frac{N}{\sum_{i=1}^N \text{Corr}(w, w'_i)} - \text{Corr}(X, X^w) \quad (12)$$

$$f = \frac{N}{\sum_{i=1}^N \text{NC}(w, w'_i)} - \text{NC}(X, X^w) \quad (13)$$

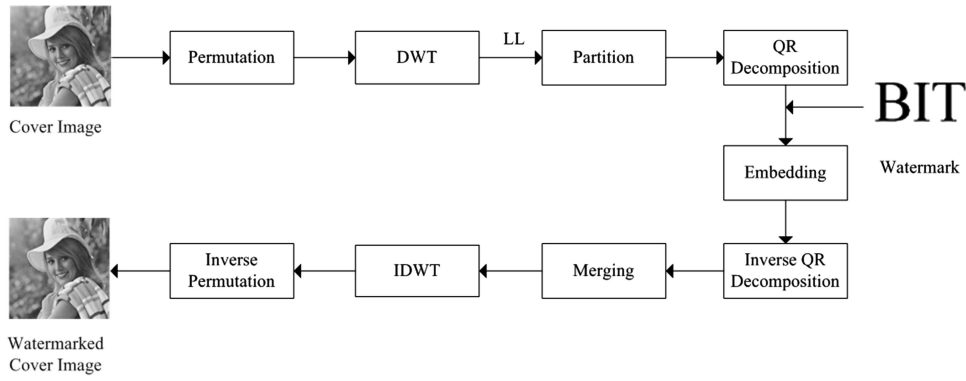


Fig. 1 Procedure of watermark embedding

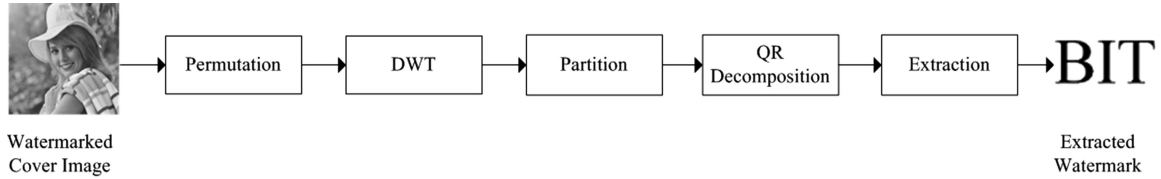


Fig. 2 Procedure of watermark extraction

$$f = 10 \cdot [1 - \text{SSIM}(X, X^w)] + \frac{1}{N} \sum_{i=1}^N \text{BER}(w, w'_i) \quad (14)$$

The objective function used in the proposed optimised image watermarking algorithm is the combination of SSIM and BER as given by.

$$f = [1 - \text{SSIM}(X, X^w)] + \lambda \cdot \frac{1}{N} \sum_{i=1}^N \text{BER}(w, w'_i) \quad (15)$$

where  $\lambda$  is the weight factor.  $X$  and  $X^w$  denote original cover image and watermarked cover image respectively.  $w$  is original watermark,  $w'_i$  indicates extracted watermark under  $i$ th attack.  $N$  represents the total number of attacks.

There are two reasons to prefer the objective function depend on SSIM and BER: (i) the objective function used in [4] is given by (14) and it has been proved in [4] that the SSIM as an image quality measure index is superior to the traditional PSNR; (ii) as an index to measure robustness, BER can describe the error more intuitively than NC and SSIM.

Although the proposed objective function given by (15) is similar to (14) used in [4], it is necessary to point out that the weights in the proposed objective function are different from (14). The objective function given by (14) is designed to focus on watermark invisibility, whereas the proposed objective function is primarily focus on watermark robustness. In this paper, we select  $\lambda = 30$  as the weight factor according to following reasons: (i) the proposed method is primarily focus on watermark robustness, the weight of watermark robustness should larger than that of watermark invisibility; (ii) in order to meet the need of watermark invisibility (i.e.  $\text{SSIM}(X, X^w) > 0.9$ ), the weight of watermark robustness should less than some value. For a different data sets, according to the above theoretical analysis, the weight factor can be slightly adjusted to meet the different requirements of needs. In Section 4.1, it will be shown that the proposed method can not only obtain high robustness, but also meet the need of watermark invisibility.

The detailed procedure of FA-based watermarking method in DWT-QR transform domain are introduced as follows and the flow chart of this is shown in Fig. 3:

Step 1: Initialise the basic parameters of FA and generate randomly the locations  $\lambda_i (i = 1, 2, \dots, n)$  of fireflies  $i = 1, 2, \dots, n$ .

Step 2: For the location  $\lambda_i$  of each firefly  $i$ , the operations are as follows:

- According to the procedure of watermarking embedding shown in Fig. 1, the watermarked cover image  $X^w$  can be obtained using the cover image  $X$  and watermark embedding strength  $\lambda_i$ .
- Apply  $N$  different attacks on the watermarked cover images, respectively. Based on above images, the extracted watermarks  $w'_i$  can be obtained by using the procedure of watermarking extraction shown in Fig. 2.
- Compute  $\text{SSIM}(X, X^w)$  and  $\text{BER}(w, w'_i)$  based on the results of i and ii.
- Calculate the objective function value of each  $\lambda_i$  using the formula as follows:

$$f(\lambda_i) = [1 - \text{SSIM}(X, X^w)] + 30 \cdot \frac{1}{N} \sum_{i=1}^N \text{BER}(w, w'_i) \quad (16)$$

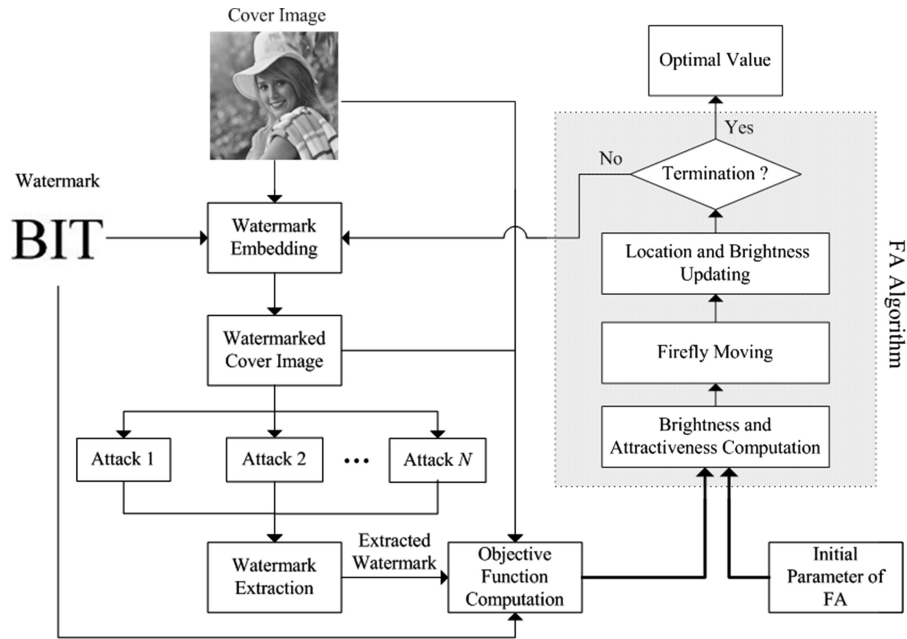
Step 3: Update the location of each firefly according to (3).

Step 4: Repeat steps 2 and 3 until the maximum iteration  $T$  is reached. The optimal watermark embedding strength is output as the finally result.

## 4 Experimental results

In this section, the invisibility and robustness of the proposed image watermarking method is analysed by simulation experiments. All simulation experiments are conducted on a personal computer having Intel dual core 3.2 GHz CPU with 4.0 GB RAM, and using MATLAB version R2014a under the Windows 7 environment. Meanwhile, two normal  $512 \times 512$  images shown in Figs. 4a and b are used as the cover images. A  $64 \times 64$  'BIT' image is used as watermark image, as shown in Fig. 4c. Consider the reasons shown in Section 2.3 and the fairness of the comparison with [38], the parameters of FA are selected identical to [38], i.e.  $\alpha = 0.01$ ,  $\beta_0 = \gamma = 1$ ,  $n = 10$ ,  $T = 10$ . The brightness of each firefly equal to the objective function given by (15). The results of the proposed method are compared with the methods given in [10, 20, 25, 26, 38], where the methods given in [25, 26, 38] are the optimised non-blind image watermarking methods and the methods used in [10, 20] are the traditional blind image watermarking methods without using any optimisation algorithm.

The performance of the proposed algorithm is shown in the following two subsections. In Section 4.1, the invisibility is detected by subjective visual observation and objective data



**Fig. 3** Flow chat of FA-based watermarking method in DWT-QR transform domain



**Fig. 4** Test cover images and watermark  
(a) Lena, (b) Elaine, (c) BIT

analysis. The values of NC, SSIM, PSNR and Corr are obtained by optimising the objective function. Two different types of attacks are used, i.e. the identical attack with different parameters as described in Table 1 and the different attacks described in Table 2. The Tables 1 and 2 give every attack with a unique attack index. In Section 4.2, different attacks (index 1–10) described in Table 2 are selected as an example to detect the robustness of the proposed method. Different attacks applied to Lena are shown in Fig. 5.

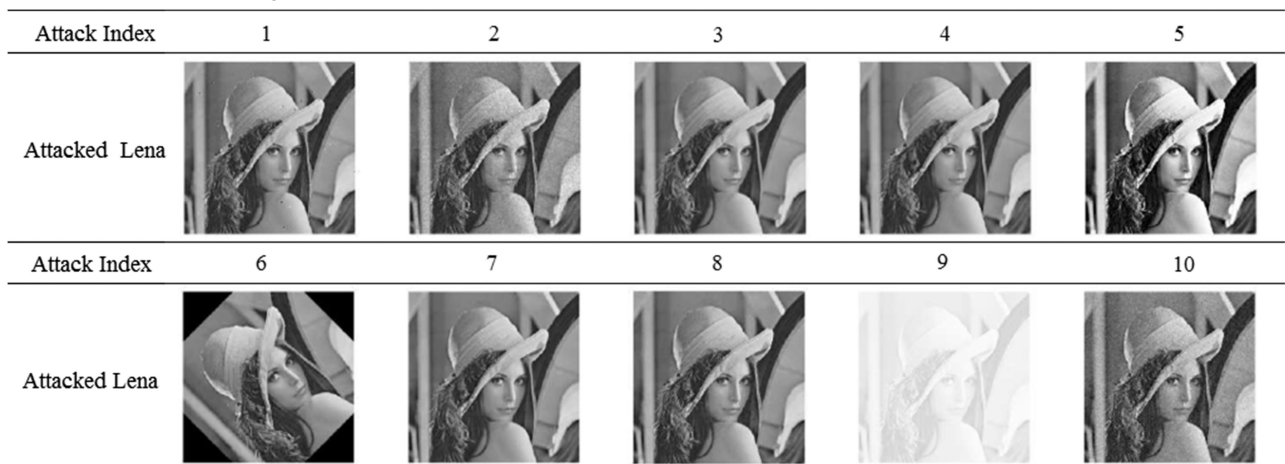
#### 4.1 Invisibility of the proposed method

The embedded watermark should be invisible to human eyes in order to ensure the information safety, hence watermark invisibility is an important index to measure the power of method.

Based on the consideration of the above facts, the watermark is embedded into two cover images and the watermarked cover

images are attacked by various attacks as given in Tables 1 and 2. The optimal watermark embedding strength is obtained by FA. Based on these optimal values, the corresponding watermarked cover images with different attack indexes are listed in Figs. 6 and 7, respectively. It can be seen from Figs. 6 and 7 that there is no visual quality difference between the watermarked and original cover images.

In this paper, the SSIM value between the original cover image and watermarked cover image is used in the objective function for measurement of invisibility. Therefore, based on these optimal watermark embedding strength, the corresponding SSIM values are calculated. Moreover, in order to prove the effectiveness of SSIM in the measurement of invisibility, the performance metrics including PSNR, NC and Corr are also calculated, respectively. All these performance metrics values are listed in the Tables 3 and 4 for Lena and Elaine, respectively. By examining Tables 3 and 4, it has been observed that all PSNR values and mean are >35 dB. Generally, if PSNR value is >35 dB, the watermarked image is within the acceptable degradation levels and thus the watermark is almost invisible to human visual system [3]. All these NC, Corr and SSIM values are >0.93. The closer of NC and Corr to 1, the higher similarity between the original cover image and watermarked cover images. Hence the proposed method meets the need of watermark invisibility for both subjective visual observation and objective data analysis.



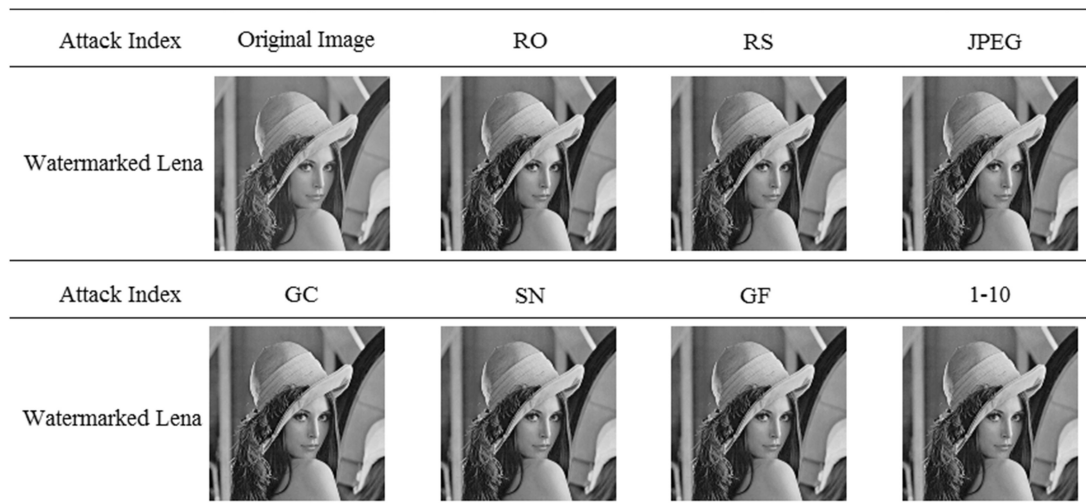
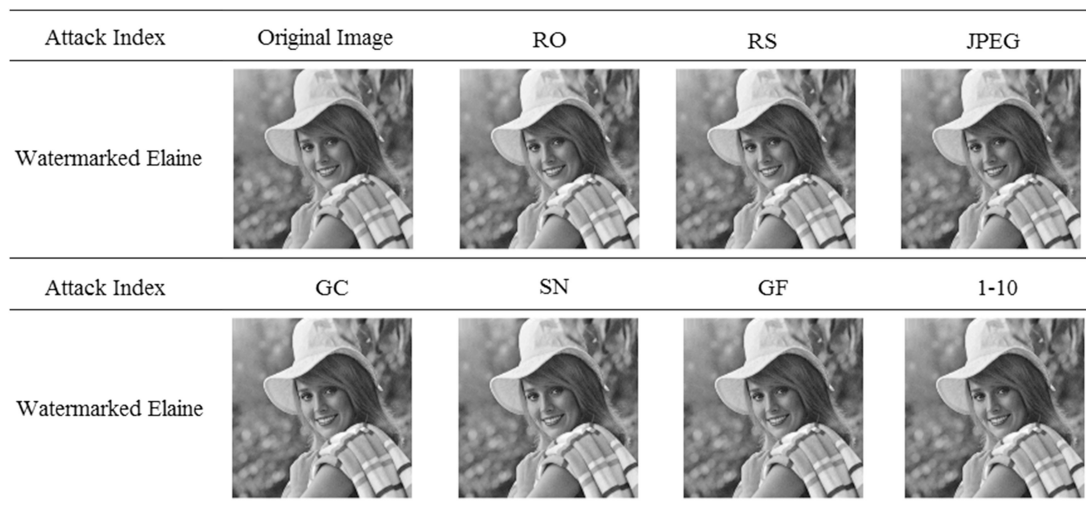
**Fig. 5** Attacked Lena with different attack indexes

**Table 1** Summary of identical attack with different parameters

| Attack index | Parameters of identical attack                                                                     |
|--------------|----------------------------------------------------------------------------------------------------|
| RO           | counter clockwise rotation with 15°, 30°, 45°, 60°, 75°, 90°                                       |
| RS           | rescaling with the factor 0.5, 0.6, 0.7, 0.8, 0.9, 1.0                                             |
| JPEG         | JPEG compression with quality factor 30, 35, 40, 45, 50, 55                                        |
| GC           | gamma correction with gamma value 0.1, 0.2, 0.3, 0.4, 0.5, 0.6                                     |
| SN           | speckle noise with variance 0.001, 0.002, 0.003, 0.004, 0.005, 0.006                               |
| GF           | Gaussian low-pass filter of size $3 \times 3$ with standard deviation 0.6, 1.2, 1.8, 2.4, 3.0, 3.6 |

**Table 2** Summary of different attacks

| Attack index | Description                                                             |
|--------------|-------------------------------------------------------------------------|
| 1            | counter clockwise rotation with 45°                                     |
| 2            | rescaling $512 \rightarrow 256 \rightarrow 512$                         |
| 3            | JPEG compression with quality factor 25                                 |
| 4            | gamma correction with gamma value 0.1                                   |
| 5            | Gaussian noise with mean zero and standard deviation 0.002              |
| 6            | salt and pepper noise with noise density 0.001                          |
| 7            | speckle noise with variance 0.005                                       |
| 8            | Gaussian low-pass filter of size $3 \times 3$ with standard deviation 3 |
| 9            | median filter with window size $4 \times 4$                             |
| 10           | histogram equalisation                                                  |

**Fig. 6** Original and watermarked Lena with different attack indexes**Fig. 7** Original and watermarked Elaine with different attack indexes

#### 4.2 Robustness of the proposed method

Robustness is defined as ‘the ability of a system to resist change without adapting its initial stable configuration’. In image watermarking, robustness means that the ability to extract clear watermark from the watermarked cover image under various attacks. Therefore, it is important to detect the robustness of an image watermarking method. To detect the performance of the proposed method, this subsection compares the robustness of the proposed method with other five methods [10, 20, 25, 26, 38]. The extracted watermarks are shown in Figs. 8 and 9. The quality of the

extracted watermark is measured by BER, NC and SSIM, the corresponding results are listed in Tables 5–7, respectively.

By examining Figs. 8 and 9, the visual quality of extracted watermark based on the proposed method is comparable to the methods given by [25, 26] for all attacks especially for gamma correlation (GC). For GC (index 4), the extracted watermark based on the proposed method is clear, however the extracted watermark based on three methods given in [25, 26, 38] are blurry, especially for [38]. Compared with the methods given in [10, 20], the visual quality of extracted watermark based on the proposed method is better, especially for GC (index 4) and histogram equalisation

**Table 3** Results of four metrics for Lena

| Attack index | SSIM   | PSNR    | NC     | Corr   |
|--------------|--------|---------|--------|--------|
| RO           | 0.9629 | 40.4930 | 0.9998 | 0.9988 |
| RS           | 0.9336 | 37.7711 | 0.9997 | 0.9978 |
| JPEG         | 0.9314 | 37.6117 | 0.9997 | 0.9978 |
| GC           | 0.9420 | 38.4161 | 0.9998 | 0.9981 |
| SN           | 0.9260 | 37.2488 | 0.9997 | 0.9976 |
| GF           | 0.9314 | 37.6157 | 0.9997 | 0.9978 |
| 1–10         | 0.9190 | 36.8125 | 0.9996 | 0.9973 |
| Mean         | 0.9352 | 37.9956 | 0.9997 | 0.9979 |

**Table 4** Results of four metrics for Elaine

| Attack index | SSIM          | PSNR           | NC            | Corr          |
|--------------|---------------|----------------|---------------|---------------|
| RO           | 0.9692        | 40.2484        | 0.9999        | 0.9987        |
| RS           | 0.9420        | 37.3343        | 0.9997        | 0.9975        |
| JPEG         | 0.9204        | 35.8341        | 0.9996        | 0.9964        |
| GC           | 0.9522        | 38.234         | 0.9998        | 0.9979        |
| SN           | 0.9565        | 38.6711        | 0.9998        | 0.9981        |
| GF           | 0.9208        | 35.8573        | 0.9996        | 0.9965        |
| 1–10         | 0.9281        | 36.324         | 0.9997        | 0.9968        |
| Mean         | <b>0.9413</b> | <b>37.5005</b> | <b>0.9997</b> | <b>0.9974</b> |

Bold values indicate the advantage of this method for index 4.

(index 10) attacks. For these attacks, the extracted watermark based

| Attack Index | 1                                                                                   | 2                                                                                  | 3                                                                                  | 4                                                                                   | 5                                                                                  | 6   | 7   | 8                                                                                     | 9                                                                                     | 10                                                                                    |
|--------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----|-----|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Ours         | BIT                                                                                 | BIT                                                                                | BIT                                                                                | BIT                                                                                 | BIT                                                                                | BIT | BIT | BIT                                                                                   | BIT                                                                                   | BIT                                                                                   |
| [25]         | BIT                                                                                 | BIT                                                                                | BIT                                                                                |    | BIT                                                                                | BIT | BIT | BIT                                                                                   | BIT                                                                                   | BIT                                                                                   |
| [26]         | BIT                                                                                 | BIT                                                                                | BIT                                                                                |    | BIT                                                                                | BIT | BIT | BIT                                                                                   | BIT                                                                                   | BIT                                                                                   |
| [20]         |    | BIT                                                                                | BIT                                                                                |    |   | BIT | BIT |    |    |    |
| [10]         |   |  |  |   |  | BIT | BIT |   |   |   |
| [38]         |  | BIT                                                                                | BIT                                                                                |  | BIT                                                                                | BIT | BIT |  |  |  |

**Fig. 8** Extracted watermarks via using different methods for Lena

| Attack Index | 1                                                                                   | 2                                                                                   | 3                                                                                   | 4                                                                                   | 5                                                                                   | 6   | 7   | 8                                                                                     | 9                                                                                     | 10                                                                                    |
|--------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----|-----|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Ours         | BIT                                                                                 | BIT                                                                                 | BIT                                                                                 | BIT                                                                                 | BIT                                                                                 | BIT | BIT | BIT                                                                                   | BIT                                                                                   | BIT                                                                                   |
| [25]         | BIT                                                                                 | BIT                                                                                 | BIT                                                                                 |  | BIT                                                                                 | BIT | BIT | BIT                                                                                   | BIT                                                                                   | BIT                                                                                   |
| [26]         | BIT                                                                                 | BIT                                                                                 | BIT                                                                                 |  | BIT                                                                                 | BIT | BIT | BIT                                                                                   | BIT                                                                                   | BIT                                                                                   |
| [20]         |  | BIT                                                                                 | BIT                                                                                 |  |  | BIT | BIT |  |  |  |
| [10]         |  |  |  |  |  | BIT | BIT |  |  |  |
| [38]         |  | BIT                                                                                 | BIT                                                                                 |  | BIT                                                                                 | BIT | BIT |  |  |  |

**Fig. 9** Extracted watermarks via using different methods for Elaine

on the method in [10, 20] are too blurry to recognise. In conclusion, the proposed method is robust against all the attacks given in Table 2, especially the robustness for GC is outstanding.

The following conclusions are made from the analysis of these experimental datas listed in Tables 5–7:

- In comparison with the optimised image watermarking methods given in [25, 26], the performance of the proposed method is better than or comparable with [25, 26] for the robustness of rotation, Gaussian low-pass filter, median filter, scaling, GC and JPEG compression. It is worth noting that the

**Table 5** BER between the original and extracted watermarks

| Image  | Index | Ours     | [25]          | [26]          | [20]          | [10]          | [38]          |
|--------|-------|----------|---------------|---------------|---------------|---------------|---------------|
| Lena   | 1     | 0.0012   | 0.0063        | 0.0063        | 0.3413        | 0.4045        | 0.4658        |
|        | 2     | 0        | 0             | 0             | 0.1084        | 0.4531        | 0             |
|        | 3     | 0        | 0             | 0             | 0.2144        | 0.4116        | 0.0037        |
|        | 4     | <b>0</b> | <b>0.8523</b> | <b>0.8523</b> | <b>0.7988</b> | <b>0.8394</b> | <b>0.7778</b> |
|        | 5     | 0.0142   | 0             | 0             | 0.5259        | 0.5688        | 0.0166        |
|        | 6     | 0.0044   | 0             | 0             | 0.0488        | 0.0378        | 0             |
|        | 7     | 0.0076   | 0             | 0             | 0.3210        | 0.3328        | 0.0039        |
|        | 8     | 0        | 0.0049        | 0.0015        | 0.2842        | 0.5610        | 0.0247        |
|        | 9     | 0        | 0.0015        | 0.0005        | 0.6011        | 0.4990        | 0.1960        |
|        | 10    | 0.0156   | 0.0017        | 0.0015        | 0.8928        | 0.9526        | 0.1267        |
| Elaine | 1     | 0.0012   | 0.0056        | 0.0056        | 0.3083        | 0.3450        | 0.4695        |
|        | 2     | 0        | 0             | 0             | 0.0381        | 0.4207        | 0             |
|        | 3     | 0        | 0             | 0             | 0.2156        | 0.3806        | 0.0059        |
|        | 4     | <b>0</b> | <b>0.8518</b> | <b>0.8518</b> | <b>0.8745</b> | <b>0.8879</b> | <b>0.8167</b> |
|        | 5     | 0.0010   | 0             | 0             | 0.5496        | 0.5361        | 0.0168        |
|        | 6     | 0.0029   | 0             | 0             | 0.0476        | 0.0349        | 0             |
|        | 7     | 0.0005   | 0             | 0             | 0.3909        | 0.3706        | 0.0125        |
|        | 8     | 0        | 0.0027        | 0.0017        | 0.1868        | 0.5500        | 0.0266        |
|        | 9     | 0        | 0.0039        | 0.0022        | 0.5884        | 0.5049        | 0.2070        |
|        | 10    | 0.0159   | 0.0020        | 0.0022        | 1.0776        | 0.6782        | 0.1829        |

Bold values indicate the advantage of this method for index 4.

**Table 6** NC between the original and extracted watermarks

| Image  | Index | Ours          | [25]          | [26]          | [20]          | [10]          | [38]          |
|--------|-------|---------------|---------------|---------------|---------------|---------------|---------------|
| Lena   | 1     | 0.9993        | 0.9964        | 0.9964        | 0.8930        | 0.8718        | 0.6889        |
|        | 2     | 1.0000        | 1.0000        | 1.0000        | 0.9673        | 0.8558        | 1.0000        |
|        | 3     | 1.0000        | 1.0000        | 1.0000        | 0.9344        | 0.8700        | 0.9979        |
|        | 4     | <b>1.0000</b> | <b>0.1771</b> | <b>0.1771</b> | <b>0.7287</b> | <b>0.7126</b> | <b>0.3406</b> |
|        | 5     | 0.9920        | 1.0000        | 1.0000        | 0.8303        | 0.8151        | 0.9905        |
|        | 6     | 0.9975        | 1.0000        | 1.0000        | 0.9854        | 0.9887        | 1.0000        |
|        | 7     | 0.9957        | 1.0000        | 1.0000        | 0.9000        | 0.8962        | 0.9978        |
|        | 8     | 1.0000        | 0.9972        | 0.9992        | 0.9120        | 0.8179        | 0.9859        |
|        | 9     | 1.0000        | 0.9992        | 0.9997        | 0.8033        | 0.8399        | 0.8816        |
|        | 10    | 0.9912        | 0.9990        | 0.9992        | 0.6906        | 0.6656        | 0.9257        |
| Elaine | 1     | 0.9993        | 0.9968        | 0.9968        | 0.9039        | 0.8919        | 0.6870        |
|        | 2     | 1.0000        | 1.0000        | 1.0000        | 0.9886        | 0.8671        | 1.0000        |
|        | 3     | 1.0000        | 1.0000        | 1.0000        | 0.9340        | 0.8803        | 0.9967        |
|        | 4     | <b>1.0000</b> | <b>0.1786</b> | <b>0.1786</b> | <b>0.6980</b> | <b>0.6923</b> | <b>0.2681</b> |
|        | 5     | 0.9994        | 1.0000        | 1.0000        | 0.8218        | 0.8267        | 0.9904        |
|        | 6     | 0.9983        | 1.0000        | 1.0000        | 0.9858        | 0.9896        | 1.0000        |
|        | 7     | 0.9997        | 1.0000        | 1.0000        | 0.8769        | 0.8837        | 0.9929        |
|        | 8     | 1.0000        | 0.9985        | 0.999         | 0.9430        | 0.8222        | 0.9848        |
|        | 9     | 1.0000        | 0.9978        | 0.9988        | 0.8080        | 0.8380        | 0.8745        |
|        | 10    | 0.9910        | 0.9989        | 0.9988        | 0.6062        | 0.7793        | 0.8910        |

Bold values indicate the advantage of this method for index 4.

GC robustness of the proposed method is outstanding. For Gaussian noise, speckle noise, salt, and pepper noise and histogram equalisation, the proposed method is slightly worse than the two methods [25, 26]. Compared with the optimised image watermarking method given in [38], the performance of the proposed method is better than the [38] for the robustness of most attacks, especially for rotation, GC, Gaussian low-pass filter and histogram equalisation.

- ii. In comparison with the two methods [10, 20], the proposed method surpasses in terms of robustness of all attacks, especially for GC and histogram equalisation attacks. The results obtained above are also consistent with the visual observation.

## 5 Conclusions

In this paper, a novel optimised image watermarking method based on FA in DWT-QR domain is proposed. The invisibility and robustness of this method are detected by simulation experiment for two cover images. The watermarked cover images have good visual quality with high SSIM, NC, Corr and PSNR values. The watermarks can be extracted from the watermarked cover image under different attacks listed in Table 2. The analysis of these experimental results show that the proposed image watermarking method not only meets the need for the invisibility, but also robust against to some image processing attacks and geometric attacks. In comparison with the three optimised image watermarking methods given in [25, 26, 38], the performance metric values including NC, SSIM and BER have reflected that the proposed method has better or comparable performance in term of robustness for most of the attacks. Compared with the methods given in [10, 20], the proposed method outperforms the traditional blind image

**Table 7** SSIM between the original and extracted watermarks

| Image  | Index | Ours          | [25]          | [26]          | [20]          | [10]          | [38]          |
|--------|-------|---------------|---------------|---------------|---------------|---------------|---------------|
| Lena   | 1     | 0.9992        | 0.9032        | 0.9212        | 0.3329        | 0.2368        | 0.0519        |
|        | 2     | 1.0000        | 1.0000        | 1.0000        | 0.4213        | 0.1296        | 1.0000        |
|        | 3     | 1.0000        | 1.0000        | 1.0000        | 0.2384        | 0.1464        | 0.9043        |
|        | 4     | <b>1.0000</b> | <b>0.0044</b> | <b>0.0044</b> | <b>0.0226</b> | <b>0.0128</b> | <b>0.0079</b> |
|        | 5     | 0.6900        | 1.0000        | 1.0000        | 0.1201        | 0.1115        | 0.6128        |
|        | 6     | 0.9126        | 1.0000        | 1.0000        | 0.5124        | 0.6802        | 1.0000        |
|        | 7     | 0.8561        | 1.0000        | 1.0000        | 0.2112        | 0.1883        | 0.8775        |
|        | 8     | 1.0000        | 0.8568        | 0.9888        | 0.2397        | 0.1035        | 0.7379        |
|        | 9     | 1.0000        | 0.9752        | 0.9951        | 0.0977        | 0.1076        | 0.2305        |
|        | 10    | 0.9965        | 0.9892        | 0.9947        | 0.0174        | 0.0059        | 0.3135        |
| Elaine | 1     | 0.9992        | 0.9675        | 0.9675        | 0.5333        | 0.3217        | 0.0314        |
|        | 2     | 1.0000        | 1.0000        | 1.0000        | 0.6032        | 0.1428        | 1.0000        |
|        | 3     | 1.0000        | 1.0000        | 1.0000        | 0.2391        | 0.1578        | 0.8203        |
|        | 4     | <b>1.0000</b> | <b>0.0044</b> | <b>0.0044</b> | <b>0.0227</b> | <b>0.0266</b> | <b>0.0053</b> |
|        | 5     | 0.9695        | 1.0000        | 1.0000        | 0.1174        | 0.1177        | 0.6154        |
|        | 6     | 0.935         | 1.0000        | 1.0000        | 0.5023        | 0.6762        | 1.0000        |
|        | 7     | 0.9944        | 1.0000        | 1.0000        | 0.1711        | 0.1647        | 0.6710        |
|        | 8     | 1.0000        | 0.9113        | 0.9208        | 0.2987        | 0.1061        | 0.7398        |
|        | 9     | 1.0000        | 0.8579        | 0.9279        | 0.0963        | 0.1093        | 0.2252        |
|        | 10    | 0.9965        | 0.9939        | 0.9929        | 0.0116        | 0.0013        | 0.2644        |

Bold values indicate the advantage of this method for index 4.

watermarking without using optimisation algorithm for all attacks. It is worth noting that the GC robustness of proposed method is outstanding.

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